

Anthony M. Alam

Ottawa, Ontario, Canada | Open to relocating within Canada and to the U.S.

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Software engineer with production experience across backend infrastructure, test automation, and full-stack development, from an internal Cisco tool adopted by 75+ engineers to a self-shipped iOS app with 20 daily active users on the App Store.

EDUCATION

University of Ottawa

Bachelor of Applied Science in Software Engineering - GPA: 3.96/4.00

Ottawa, Ontario

Graduation: December 2027

EXPERIENCE

Cisco Systems

Software Engineer Intern (Telemetry and Programmability)

Ottawa, Ontario

May 2026 - August 2026

- Owned the development of a real-time observability service end to end, eliminating a telemetry message drop blind spot by mapping failures to 15+ actionable root causes across NETCONF and gRPC subscriptions.
- Implemented public IOS XE CLIs and extended the corresponding YANG model to expose the new drop cause data, backed by automated unit tests and projected to save 100+ engineering hours during its first year in production.
- Designed a Python (pyATS) analysis tool to quantify the memory overhead telemetry subscriptions impose, introducing a novel Memory Multiplier heuristic and enabling a 10x reduction in memory overhead via optimized sizing.

Software Engineer Intern (Code Injection and Infrastructure)

January 2025 - April 2025

- Spearheaded the development of a Python (Flask) internal code injection tool adopted by 75+ engineers, allowing previously unreachable conditions to be tested and increasing file line and branch coverage by 22% and 13% on average.
- Architected and re-implemented an optimal, scalable ConditionBlock graph data structure to support DAG-based flow analysis of C code, improving run time by 18% and cutting memory usage by 20%.
- Developed a library of 15 reusable APIs and algorithms for ConditionBlock graphs, already leveraged across 20+ Python files to support graph traversals, conditional path generation, and SQL-based data persistence (SQLite).
- Designed and implemented automated pre-commit routines with Python (pyATS) and YAML, covering all known edge cases and eliminating a 2-hour manual upgrade/downgrade process for Cisco Catalyst high-availability switches.

Software Engineer Intern (Test Automation)

May 2024 - August 2024

- Led the development of an automated Python (pyATS) test suite covering 50+ gRPC/gNMI request scenarios for a multimillion-dollar customer deployment.
- Increased line coverage by 5% and branch coverage by 4% by writing unit tests in C (Cmocka), helping the team reach its 80% coverage goal.

Ford Motor Company

Software Developer Intern (Vehicle Wi-Fi)

Ottawa, Ontario

September 2025 - December 2025

- Spearheaded the development of a production C++ service-oriented architecture (SOA) proxy service for Ford's in-vehicle software platform, enabling hotspot credential read/write operations across 400,000+ next-generation Ford and Lincoln vehicles.

PROJECTS

PlayItAgain: Kolonie - iOS App | *Swift, SwiftUI, YOLOv8, Core ML*

[App Store](#) | [Support Page](#)

- Shipped a production iOS app in Swift/SwiftUI that digitizes physical Catan boards and game state from a single photo, letting players save and resume games anytime, surpassing 100+ downloads on the App Store.
- Engineered a Core Image perspective-correction pipeline and on-device YOLOv8 + Core ML detector, achieving 98% board detection accuracy and sustaining 20 daily active users.

Grant-Winning Accessibility Tool (for Rutgers University) | *JavaScript, HTML, CSS, OpenAI API*

[GitHub](#)

- Spearheaded the development of a full-stack web chatbot that accurately evaluates accessibility across 300+ Rutgers University online courses, earning a grant for AI innovation from the University of Ottawa.
- Built a custom CSV parsing algorithm to process 3000+ course components in under 1 second, enabling faculty to upload course content and instantly receive tailored accessibility recommendations based on WCAG 2.1 guidelines.

TECHNICAL SKILLS

Programming & Scripting Languages: Python, Java, C/C++, Swift, JavaScript, SQL, Bash, YAML, HTML, CSS

Frameworks, Cloud & Tools: AWS, Docker, REST API, Flask, SwiftUI, PostgreSQL, NoSQL, SQLite, pyATS, JUnit, Cmocka, Git, GitHub, CI/CD, Jenkins, Unix/Linux (RHEL), VS Code, IntelliJ, Figma, Xcode, Android Studio, Claude Code, GitHub Copilot, OpenAI Codex

Engineering Skills: Full-Stack Development, Distributed Systems, Automation Testing, Monitoring & Observability, OOP, Networking (HTTP, IP, TCP, UDP), SDLC, Agile/Scrum, Cloud Computing, Debugging (GDB), Code Review, Complexity Analysis

Relevant Coursework: Data Structures & Algorithms, Software Engineering Design Patterns, Operating Systems